

Tribhuvan University
Faculty of Management
M. Phil. in Public Administration, 2079

First Semester Examination
MPH 502: Research Methods in Public Administration-II

Full Marks: 100
Time: 4 hours

Read the instruction and answer accordingly

This exam paper is divided into two parts. Part A is theory-based and Part B computer based.

Unless and otherwise stated, all notations/ terminologies have their usual meanings

Answer sheet of both part must be submitted

Part “A”

[Full Marks: 30]

[time: two hours]

Attempt any five questions

1. A restaurant owner residing at Thamel wants to study whether there is any association between amounts of wine consumed and the country of origin of the tourist. The following are the information regarding the amount of wine consumed and country of origin.

Amount Consumed	England	France	Italy	Sweden	US
Never drink	20	10	15	8	12
One glass or less	72	77	70	62	68
Between one and two	85	65	95	95	48
More than two	85	79	77	85	79

Use appropriate to test where there is a dependency between wine consumption and country of origin.

2. A study has been done on how much time an adult indulge in Facebook. A researcher conducted a survey of 50 people and found that the average amount of time spent on Facebook was 6.5 hours per week with a standard deviation of 1.5 hours. They also found that the time spent in Facebook follows a normal distribution.

Construct a 98% confidence interval for the true average time spent in Facebook for all people.

3. The citizen report cards (CRC) indicates that of 200 citizens of Kathmandu Metropolitan city showed 90 were satisfied whereas in the case of Lalitupur 120 citizens were satisfied out of 200 surveyed. Test the hypothesis that the rate of satisfaction in these two metropolitan cities are significantly different.
4. What do you mean by sampling? If you have to do a survey on tourist, some with different countries, which sampling technique you will prefer and why?

5. Elaborate the differences among the regression analysis, factor analysis, and cluster analysis?

6. The following are the sales figure before and after advertisement is given as:

Before	75	70	53	77	85	97	103	105
After	77	75	58	80	89	100	110	109

Test whether advertisement was effective.

7. A previous survey indicated that the average smart phone user earns Rs 355,600 per year. Suppose an analyst believes that the average annual earnings of a smart phone user are lower now, and he sets up a study in an attempt to prove his beliefs. He randomly samples 36 smart phone users and finds out that the average annual salary for this sample is Rs 338,974 with a population standard deviation of Rs 7,810. Use $\alpha = .01$ to test the analyst's belief. Assume wages of smart phone users are normally distributed in the population.

Part “B”**[Full Marks: 30]****[time: two hours]**

Read the instruction and answer accordingly

Unless and otherwise stated, all notations/ terminologies have their usual meanings

Answer sheet of both part must be submitted

All questions are compulsory and each carries 15 marks

1. A survey has done by the school of public administration with 20 people and the following information was found. The data is attached with this file with filename: surveydata.sav

SN	Gender	age	Income	height	BPB	Occupation	BPA
1	Male	34	555000	5.9	120	Business	110.00
2	Male	42	50000	5.5	120	Civil Service	110.00
3	Male	40	40000	6.8	120	Civil Service	100.00
4	Male	30	200000	5.5	80	Business	80.00
5	Male	31	60000	5.3	120	Civil Service	110.00
6	Male	28	12000	5.5	120	Teacher	105.00
7	Male	36	6000000	5.8	120	Bankers	105.00
8	Male	33	500000	5.8	110	Civil Service	105.00
9	Female	26	42989	5.1	130	Teacher	120.00
10	Male	45	50000	5.8	130	Teacher	115.00
11	Male	42	35000	5.5	120	Civil Service	115.00
12	Male	45	500000	5.2	120	Bankers	110.00
13	Female	26	120000	5.2	120	Business	105.00
14	Male	40	50000	5.8	120	Civil Service	110.00
15	Male	45	400000	5.3	130	Business	110.00
16	Male	39	50000	6.9	140	Teacher	120.00
17	Male	50	500000	6.1	120	Civil Service	110.00
18	Male	55	240000	5.8	110	Business	110.00
19	Female	52	280000	5.2	125	Civil Service	110.00
20	Female	47	425000	5.4	136	Bankers	110.00

Use the tab analyse in SPSS to give the answer to the following questions

- a) Find the average income of male and female with standard deviations and counts

Income			
Sex	N	Mean	Standard Deviation
Male			
Female			

b) Is the income significantly different between male and female?

Value of test statistics	P-value	Conclusion

a) Find the average income, standard deviation and count of different level of occupation.

Income			
Level of occupation	N	Mean	Standard Deviation
Total			

a) Use appropriate test statistics to test whether the different level have different income.

Value of test statistics	P-value	Conclusion

a) If there is difference then which pair has significantly different average value?

9. The quantity sold of bread is thought to be related to both price and advertisement expenses
For a random sample of 11 sales, the following data were obtained:

Quantity sold x1	Price (Rs) x2	Advertisement expenses (Rs 000) x3
132	52	173
143	59	184
153	67	194
162	73	211
154	64	196
168	74	220
166	72	217
128	46	167
159	65	207
149	61	188
137	54	188

The data is given to you with the file name sell.sav. On the basis of this data, operate the different function in SPSS and answer the followings:

- i) Generate summary statistics, including the mean and standard deviation of each variable. Compute the coefficient of variation for each variable. Relative to its mean, which variable has the greatest and smallest spread of data values?

Variable	Mean	Standard Deviation	Coefficient of variation	Comparative spread-ness
Quantity sold (x1)				
Price in Rs (x2)				
Adv expenses in Rs 000 (x3)				

- ii) For each pair of variables, generate the correlation coefficient r. Which variable (other than quantity sold) has the greatest influence on quantity sold?

Variable	Quantity Sold (x1)	Price in Rs (x2)	Adv expenses in Rs 000 (x3)
Quantity sold (x1)			
Price in Rs (x2)			
Adv expenses in Rs 000 (x3)			
Greater influencing variable to quantity sold			

- iii) Obtain the unstandardized regression coefficients of quantity sold on price and advertisement expenses along with standard error, t-value and significance. Interpret the values of the regression coefficients so obtained and also discuss the significance of these values

	Unstandardized Coefficients		t-value	Sig
	Coefficients (bi)	Standard error (sbi)		
Constant				
Price in Rs (x2)				
Adv expenses in Rs 000 (x3)				

- iv) Interpret the values of the regression coefficients so obtained and also discuss the significance of these values. Write the regression equation, find R square, adjusted R square and Standard error of estimate. Also determine what percentage of variation in quantity sold pressure is explained by this regression equation.

Regression Coefficients	Interpretation / Significance				
Regression equation					
R- square		Adjusted R-Square		Standard Error	
What percentage of variation in quantity sold is explained by this equation?					

v) Fill the ANOVA table of the model so obtained and discuss the overall model fit.

Source	df	SS	MS	F	Sig
Regression					
Residual					
Total					
Discussion on the model					